

Kalpojyoti Koch

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Profile

Computer Science graduate specializing in AI/ML and software development. I build intelligent systems using Python and PyTorch, from data pipelines to production deployment.

Looking forward to contributing to software or AI-focused product teams.

Education

Central Institute of Technology, BTech in Computer Science Aug 2021 – May 2025

- GPA: 7.33/10
- **Coursework:** Computer Architecture, DSA, Computational Theory, Database

Professional Experience

AI Research Operations Engineer, AMITY CENTRE FOR AI – Noida, Uttar Pradesh Oct 2025 – Present

- Built ML ranking & data pipelines for an ICMR-funded virus recommender system
- Worked cross-functionally to align research needs with deployable ML modules

Intern AI/ML, NIELIT National Institute of Electronics & IT – Kokrajhar, Assam Aug 2025 – Oct 2025

- Developed a cough-audio respiratory screener using CNN + spectrogram features
- Processed and refined 10k+ real-world cough samples for robust accuracy
- Optimized inference for low-compute medical/field deployment

Intern, NEEPCO – Shillong, Meghalaya Jul 2024 – Aug 2024

- Deployed a RAG policy chatbot with clause-level semantic retrieval
- Fine-tuned quantized TinyLLaMA to run on free-tier infrastructure reliably
- Implemented vector search workflows using ChromaDB

Certifications

AI & Machine learning using Python Jun 2023 - Jul 2023

National Institute of Electronics & IT

Certificate in Computer Applications May 2019 - Jul 2019

AMTRON, Assam Electronics Development Corporation

Project Experience

Automated Attendance System Using Deep Learning

- Built a real-time facial attendance system using CNN and a custom dataset of 75,000+ images
- Integrated model inference with server backend, achieving high reliability for real-world use
- Tools Used: PyTorch, SQL, ResNet50, Mediapipe

Home Automation System with Gas Detection and Mobile Control

- Designed an Arduino + GSM based automation system for remote fan/light control via mobile
- Implemented automatic ventilation triggering during gas leakage with safety threshold logic
- Tools Used: Arduino UNO, GSM module

Technical Skills

Languages: C++, C, Python, SQL

Tools & Frameworks: PyTorch, Tensorflow, LangChain, scikit-learn, FastAPI, ChromaDB, MongoDB